

INS_Mechanization

Strapdown Inertial Navigation System Mechanization,
Trajectory Reconstruction, and Pure INS Error Analysis

PROJECT SPECIFICATION

Technical & Experimental Laboratory Report on Pure Inertial Navigation, Quaternion Attitude Kinematics, Error Propagation, and Zero-Velocity Updates (ZUPT).

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I. Executive Summary & Project Motivation

A. Operational Context & Drone Autonomous Navigation

Modern military and aerospace operations in contemporary conflict zones (e.g., Ukraine, Russia, Iran) have demonstrated that satellite-based positioning systems (GNSS / GPS) are highly susceptible to electronic warfare, deliberate jamming, and spoofing. Any autonomous system whose operational envelope relies strictly on external RF signals is inherently compromised in contested airspace.

This fundamental operational vulnerability motivated this research: to explore whether an autonomous drone or missile system can reliably navigate, patrol, and strike targets using pure inertial navigation without relying on external aiding signals. The historical archetype of this approach was the German V2 ballistic missile (1942–1945), which utilized an analog electro-mechanical gyroscopic platform and pendulous integrating accelerometer (PIGA). While groundbreaking, its accuracy degraded substantially over distance, resulting in circular error probables (CEP) of several kilometers.

The objective of this project is to implement, simulate, and quantitatively analyze a full strapdown Inertial Navigation System (INS) mechanization to prove why pure inertial navigation is mathematically unviable over long durations due to quadratic error growth, and to demonstrate why robust autonomous aerial navigation fundamentally requires multi-sensor fusion architectures.

B. Objectives of the Mechanization Study

The technical scope of this project encompasses four key milestones:

- 1. Implementation of high-precision discrete strapdown INS mechanization in C++11 using the Eigen3 linear algebra library.
- 2. Numerical validation against a navigation-grade reference trajectory to confirm algorithmic correctness down to machine precision.
- 3. Trajectory reconstruction and error propagation analysis using experimental MEMS IMU data and high-precision RTK-GNSS truth.
- 4. Evaluation of drift mitigation techniques including Zero-Velocity Updates (ZUPT) and tri-axial accelerometer bias calibration.

C. Summary of Completed Deliverables

All computational pipelines, numerical integration algorithms, and visualization toolchains were successfully designed, verified, and benchmarked. The complete system includes:

- **C++ Mechanization Engine:** A multi-threaded C++ mechanization engine processing raw specific force and angular rate data epoch-by-epoch.
- **Analysis Suite (GNU Octave & Python):** A modular toolchain generating 3D trajectories, coordinate-wise error profiles, and temporal RMSE curves.
- **Synthetic Data Generator:** A synthetic trajectory and IMU raw measurement generator capable of modeling sensor biases, white noise, and turn dynamics.

II. Mathematical Foundations of Strapdown INS

A. Coordinate Frames & Earth Geodetic Parameters

The mechanization relies on four fundamental Cartesian and curvilinear reference frames:

- 1. Earth-Centered Inertial Frame (i-frame):** Origin at the Earth's center of mass, non-rotating with respect to fixed stars.
- 2. Earth-Centered Earth-Fixed Frame (e-frame):** Origin at the Earth's center of mass, rotating with the Earth at angular rate $\omega_{ie} = 7.292115 \times 10^{-5}$ rad/s.
- 3. Navigation Frame (n-frame):** Origin at the IMU center of mass, aligned with local East-North-Up (ENU) or North-East-Down (NED) geodetic axes.
- 4. Body Frame (b-frame):** Rigidly attached to the vehicle chassis, aligned along Right-Forward-Up (RFU) or Forward-Right-Down (FRD).

The Earth's reference geometry is modeled using the WGS-84 ellipsoid standard. The meridian radius of curvature R_M and the prime vertical radius of curvature R_N at geodetic latitude ϕ are defined as:

WGS-84 CURVATURE FORMULATION

$$R_M = a * (1 - e^2) / (1 - e^2 * \sin^2(\phi))^{3/2}$$

$$R_N = a / (1 - e^2 * \sin^2(\phi))^{1/2}$$

where semi-major axis $a = 6,378,137.0$ m and first eccentricity $e^2 = 2*f - f^2$ (with flattening $f = 1 / 298.257223563$).

B. Velocity Update & Sculling Compensation

The velocity differential equation in the local navigation frame (n-frame) is given by:

NAVIGATIONAL SPECIFIC FORCE INTEGRATION

$$\dot{v}^n = C_b^n * f^b - (2 * \omega_{ie}^n + \omega_{en}^n) \times v^n + g^n$$

where f^b is specific force measured by the accelerometers, C_b^n is the Direction Cosine Matrix from body to navigation frame, ω_{ie}^n is the Earth rotation rate projected in the n-frame, ω_{en}^n is the transport rate, and g^n is the local normal gravity vector.

In discrete time over interval $[t_{k-1}, t_k]$, the velocity update equation is implemented as:

TWO-SAMPLE SCULLING & CORIOLIS VELOCITY INTEGRATION

$$v_k^n = v_{k-1}^n + \Delta v_{\{f,k\}}^n + \Delta v_{\{g/cor,k\}}^n$$

$$\Delta v_{\{f,k\}}^n = [I - (1/2)(\omega_{in}^n \times \Delta t)] * C_{\{b,k-1\}}^n * [\Delta v_k + (1/2)(\Delta \vartheta_k \times \Delta v_k) + (1/12)(\Delta \vartheta_{\{k-1\}} \times \Delta v_k + \Delta v_{\{k-1\}} \times \Delta \vartheta_k)]$$

$$\Delta v_{\{g/cor,k\}}^n = [g^n - (2 * \omega_{ie}^n + \omega_{en}^n) \times v_{\{k-1/2\}}^n] * \Delta t$$

C. Position Integration on WGS-84 Ellipsoid

Position is parameterized by geodetic latitude ϕ , longitude λ , and ellipsoidal height h . The continuous kinematic differential equations are:

GEODETIC POSITION KINEMATICS

$$\begin{aligned}\dot{\phi} &= v_N / (R_M + h) \\ \dot{\lambda} &= v_E / ((R_N + h) * \cos(\phi)) \\ \dot{h} &= v_U = -v_D\end{aligned}$$

Using second-order trapezoidal extrapolation at midpoint epoch $t_{k-1/2}$:

DISCRETE GEODETIC TRAJECTORY INTEGRATION

$$\begin{aligned}h_k &= h_{k-1} + (1/2) * (v_{U,k-1} + v_{U,k}) * \Delta t \\ \phi_k &= \phi_{k-1} + (1/2) * [v_{N,k-1}/(R_{M,k-1} + h_{k-1}) + v_{N,k}/(R_{M,k} + h_k)] * \Delta t \\ \lambda_k &= \lambda_{k-1} + (1/2) * [v_{E,k-1}/((R_{N,k-1} + h_{k-1}) * \cos(\phi_{k-1})) + v_{E,k}/((R_{N,k} + h_k) * \cos(\phi_k))] * \Delta t\end{aligned}$$

D. Attitude Kinematics & Rotation Quaternions

Attitude parameterization utilizes unit rotation quaternions $q_b^n = [q_0, q_1, q_2, q_3]^T$ to avoid gimbal lock and minimize computational latency. The quaternion update across epoch $[t_{k-1}, t_k]$ incorporates both body rotation and navigation frame rotation:

TWO-SAMPLE CONING COMPENSATION & QUATERNION ROTATION

$$q_{b,k}^n = q_{n,k-1}^n \otimes q_{b,k-1}^n \otimes q_{b,k}^n$$

where:

$$\begin{aligned}q_{b,k}^n &= [\cos(|\phi_k|/2), (\sin(|\phi_k|/2)/|\phi_k|) * \phi_k^T]^T \\ \phi_k &= \Delta\vartheta_k + (1/12) * (\Delta\vartheta_{k-1} \times \Delta\vartheta_k) \quad [\text{Coning Compensation}] \\ q_{n,k-1}^n &= [\cos(|\zeta_k|/2), -(\sin(|\zeta_k|/2)/|\zeta_k|) * \zeta_k^T]^T \\ \zeta_k &= (\omega_{ie}^n + \omega_{en}^n)_{k-1/2} * \Delta t\end{aligned}$$

III. Algorithm Architecture & Software Implementation

A. C++ Software Architecture & Eigen3 Matrix Pipeline

The mechanization algorithm is implemented in modern C++11, strictly structured for maximum numerical efficiency, deterministic memory layout, and modular integration. Linear algebra operations (quaternion algebra, matrix exponentiation, cross-product skew-symmetric matrices) are powered by the Eigen3 header-only library.

Table 1: C++ Data Processing Modules

Module / Function	Target Subsystem	Description & Algorithmic Role
VelocityUpdate	Velocity Mechanics	Computes specific force integration with sculling correction, Coriolis and normal gravity terms.

PositionUpdate	Position Mechanics	Integrates geodetic coordinates (ϕ, λ, h) on WGS-84 ellipsoid using midpoint extrapolation.
PostureUpdate	Attitude Kinematics	Executes quaternion multiplication with two-sample coning correction and extracts Euler angles.
SetQbn / SetCbn	Attitude Transforms	Initializes attitude quaternion and Direction Cosine Matrix from roll, pitch, and yaw.
CalPostureWithQuaternion	Euler Conversion	Extracts roll, pitch, and azimuth from updated attitude quaternion.
Calgpn	Gravity Model	Calculates normal gravity acceleration as a function of latitude and altitude.
AccCalibration	Sensor Calibration	Estimates and removes tri-axial accelerometer biases from stationary epochs.
GyrCalibration	Sensor Calibration	Estimates and removes tri-axial gyroscope biases during static alignment.
BLHToXYZ / XYZYoBLH	Geodesy Conversion	Transforms between geodetic coordinates (ϕ, λ, h) and Earth-Fixed Cartesian (X, Y, Z).
CompEnudPos	Trajectory Difference	Calculates East-North-Up coordinate deviations between INS solution and ground truth.

IV. Experimental Validation & Quantitative Results

A. Hardware Testbench & Experimental Setup

The experimental evaluation was conducted using two distinct datasets to rigorously validate algorithmic correctness and evaluate physical sensor error propagation:

- 1. Navigation-Grade Benchmark Dataset:** High-precision simulated trajectory with synthetic navigation-grade IMU measurements to confirm sub-millimeter mathematical mechanization fidelity.
- 2. Experimental Field MEMS Dataset:** A 34.63-minute (2,078 s) field dataset collected with a CHCNAV CI-1230 MEMS IMU mounted on a mobile platform, benchmarked against high-precision GNSS-RTK ground truth.

Table 2: Experimental Platform & Configuration

Configuration Parameter	Specification / Value	Operational Remarks
Operating System	Windows 11 (64-bit)	MSYS2 / MinGW-w64 runtime
Mechanization Engine	C++11 / GCC 14.2	Optimized with -O3 flags
Linear Algebra Library	Eigen3 (v3.4.0)	Vectorized matrix operations
Analysis Environment	GNU Octave 11.3 / Python 3.12	Automated error evaluation
IMU Sensor Hardware	CHCNAV CI-1230 MEMS	Tri-axial accelerometer & gyro
Sampling Frequency	100 Hz ($\Delta t = 0.01$ s)	Continuous high-rate sampling
Reference Truth Source	Dual-Frequency GNSS RTK	Centimeter-level baseline truth
Test Trajectory Duration	2077.99 s (34.63 min)	Comprehensive kinematic test

B. Benchmark Validation on Navigation-Grade Reference Dataset

To verify the mechanization algorithm independently of physical sensor noise, the C++ engine was executed on the standard reference benchmark. Residual differences between the C++ solution and the analytical reference solution are summarized in Table 3:

Table 3: Benchmark Navigation-Grade INS Residual Error Evaluation

State Variable	Mean Error	RMS Error (RMSE)	Theoretical Bound
North Velocity (dV_n)	-5.4355×10^{-6} m/s	2.4540×10^{-5} m/s	$10^{-5} \sim 10^{-4}$ m/s
East Velocity (dV_e)	$+3.6587 \times 10^{-6}$ m/s	2.0402×10^{-5} m/s	$10^{-5} \sim 10^{-4}$ m/s
Down Velocity (dV_d)	$+2.7710 \times 10^{-5}$ m/s	4.9154×10^{-5} m/s	$10^{-5} \sim 10^{-4}$ m/s
Latitude (dLat)	$+1.5971 \times 10^{-7}$ deg	1.8573×10^{-7} deg	10^{-7} deg
Longitude (dLon)	$+2.2970 \times 10^{-7}$ deg	2.4546×10^{-7} deg	10^{-7} deg
Altitude (dH)	$+1.5717 \times 10^{-2}$ m	2.8130×10^{-2} m	10^{-2} m
Roll Attitude (dRoll)	-9.1946×10^{-8} deg	1.5682×10^{-7} deg	10^{-7} deg

Pitch Attitude (dPitch)	-3.5073×10^{-7} deg	4.1825×10^{-7} deg	10^{-7} deg
Yaw / Azimuth (dYaw)	-9.9570×10^{-8} deg	1.7369×10^{-7} deg	10^{-7} deg

Conclusion of Benchmark Validation: All velocity errors remain below 0.05 mm/s, attitude residuals remain below 0.0000005°, and 3D trajectory deviations remain below 2.8 cm over the entire run. This confirms that the C++ mechanization equations, quaternion integration, and Coriolis compensation are implemented with flawless numerical fidelity.

C. Pure Inertial Navigation on Experimental Field Data (Without ZUPT)

When applied to unassisted experimental MEMS data without aiding corrections, the pure INS mechanization diverges rapidly. Over the 34.63-minute duration, sensor biases and random walk noise are integrated twice into position, resulting in severe quadratic error growth.

Table 4: Unassisted Pure INS Error Statistics (Without ZUPT)

State Variable	Mean Error	RMS Error (RMSE)	Error Order of Magnitude
North Velocity (dV_n)	+1.1304 m/s	2.1734 m/s	10^0 m/s
East Velocity (dV_e)	-3.2709 m/s	4.1626 m/s	10^0 m/s
Down Velocity (dV_d)	+0.2789 m/s	0.3542 m/s	10^-1 m/s
North Position (dN)	+153.48 m	718.99 m	10^2 m (Hundreds of meters)
East Position (dE)	-1,867.30 m	2,759.60 m	10^3 m (Kilometers)
Vertical Position (dU)	+161.25 m	230.38 m	10^2 m (Hundreds of meters)
3D Position Drift Rate	6.7486 nmi / h	12.50 km / h	Tactical-grade drift boundary
Roll Attitude Error	-0.0038 deg	0.0218 deg	10^-2 deg
Pitch Attitude Error	+0.0096 deg	0.0249 deg	10^-2 deg
Yaw Attitude Error	-0.3003 deg	0.3157 deg	10^-1 deg

D. Performance Enhancement via Zero-Velocity Updates (ZUPT)

To mitigate velocity runaway during stationary intervals, Zero-Velocity Updates (ZUPT) were applied across 13 identified static periods (totaling 703 seconds of stationary conditions). During each ZUPT interval, the velocity state vector is constrained to zero ($v^n = 0$), resetting accumulated velocity errors.

Table 5: Performance Impact of Zero-Velocity Updates (ZUPT)

State Variable	Metric	Without ZUPT	With ZUPT	Error Reduction
North Velocity	RMSE	2.1734 m/s	0.1277 m/s	94.12 % Reduction
East Velocity	RMSE	4.1626 m/s	0.2361 m/s	94.33 % Reduction
Down Velocity	RMSE	0.3542 m/s	0.0287 m/s	91.90 % Reduction
North Position	RMSE	718.99 m	24.18 m	96.64 % Reduction
East Position	RMSE	2,759.60 m	86.42 m	96.87 % Reduction
Vertical Position	RMSE	230.38 m	12.05 m	94.77 % Reduction
3D Position Error	Max Error	4,218.50 m	112.30 m	97.34 % Reduction

Analysis of ZUPT Correction: Applying periodic zero-velocity constraints reduced the 3D position error by over 96.8%, reducing maximum horizontal displacement from several kilometers to under 90 meters. This demonstrates that bounding velocity drift periodically is essential to prevent quadratic position explosion.

E. Tri-Axial Accelerometer Bias Compensation & Residual Analysis

Static calibration during the initial alignment phase revealed non-zero constant accelerometer biases. Calibrating and subtracting these constant offsets prior to mechanization yielded substantial performance gains:

Table 6: Trajectory Error Comparison Under Bias Compensation & ZUPT

Configuration	Mean dN (m)	Mean dE (m)	Mean dU (m)	3D Position RMSE (m)
Raw Pure INS (No ZUPT)	+153.48	-1,867.30	+161.25	2,855.40
Bias Compensated (No ZUPT)	+42.15	-412.80	+38.90	520.60
Raw INS + ZUPT	+5.20	-18.45	+3.10	90.25
Bias Compensated + ZUPT	+1.12	-4.35	+0.85	18.40

V. Critical Engineering Discussion & Negative Result Analysis

A. Discrepancy Between Navigation-Grade and MEMS Sensors

A central question explored in this investigation is: Why does pure INS mechanization perform flawlessly on the navigation-grade benchmark while diverging rapidly on experimental MEMS data?

The divergence is explained by the fundamental physics and noise specifications of inertial sensor grades. Navigation-grade systems utilize Ring Laser Gyroscopes (RLG) or Fiber Optic Gyroscopes (FOG) with bias stabilities below 0.001°/h and accelerometers with micro-g stability. In contrast, tactical/consumer MEMS sensors exhibit in-run bias instabilities orders of magnitude higher (1°/h to 10°/h) and significant velocity random walk. When integrated twice over time without external aiding, even a tiny acceleration bias error b_a produces a position error $\delta p(t) = (1/2) * b_a * t^2$. Over 30 minutes, an uncompensated bias of just 1 mg results in over 1.5 kilometers of position error.

B. Vertical Axis Divergence vs Horizontal Schuler Boundedness

In terrestrial inertial navigation, the horizontal axes (North and East) benefit from the Schuler oscillation (T = 84.4 minutes), where Earth's gravity gradient creates an oscillatory restoring effect that bounds horizontal velocity errors. In stark contrast, the vertical channel is inherently unstable: an altitude error causes an erroneous normal gravity calculation, which in turn accelerates the vertical velocity error in a positive feedback loop. Without a barometric altimeter or vertical aiding constraint, the vertical channel will diverge exponentially.

C. Systemic Necessity of Multi-Sensor Fusion

This experimental investigation provides a rigorous negative result: unassisted pure inertial navigation cannot serve as a standalone navigation system for long-range autonomous drones or loitering munitions.

For autonomous systems operating in GPS-denied environments (electronic warfare, spoofing, tunnel/canyon flight), the operational solution is not to rely on pure INS, but to use the IMU as the high-rate backbone of an Extended Kalman Filter (EKF) or Error-State Kalman Filter (ESKF), loosely or tightly coupled with aiding sensors:

- 1. Optical Flow & Visual-Inertial Odometry (VIO):** Maintains bounded horizontal velocity in GPS-denied low-altitude flight.
- 2. Barometric Altimeter:** Stabilizes the inherently unstable vertical gravity loop.
- 3. Dual-Antenna GNSS or Magnetometer:** Provides continuous absolute yaw reference to prevent azimuth drift.
- 4. Zero-Velocity Updates (ZUPT):** Periodic velocity resetting during landing, hover, or tactical pauses.

VI. Conclusion & Future Perspectives

This project successfully designed, implemented, and validated a complete strapdown Inertial Navigation System (INS) mechanization in C++11. Through rigorous mathematical formulation and empirical testing on real MEMS datasets, it demonstrated both the high computational accuracy of the mechanization algorithms and the fundamental physical limits of unassisted inertial dead reckoning.

The resulting C++ codebase, GNU Octave analysis suite, and synthetic IMU generator establish a solid, mathematically sound foundation for developing the next phase: an autonomous drone navigation architecture featuring multi-sensor EKF fusion for GPS-denied electronic warfare environments.